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New insights into the India–Asia collision process from Cretaceous paleomagnetic and geochronologic results in the Lhasa terrane

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ABSTRACT

To further constrain the India–Asia collisional process, a combined paleomagnetic and geochronologic study has been carried out on the Upper Cretaceous Jingzhushan Formation redbeds and the Lower Cretaceous Dianzhong Formation volcanic rocks dated at ~121–117 Ma from the Cuoqin area in the central Lhasa terrane. Stepwise thermal demagnetization successfully isolated reliable characteristic remanent magnetization (ChRM) directions that include dual polarity and pass positive fold tests at 95% and 99% confidence levels, indicating prefolding primary magnetizations. The tilt-corrected site-mean direction for 33 redbed sites is D = 316.8°, I = 30.2° with α_{95} = 5.4°, corresponding to a paleopole at 49.0°N, 344.3°E with A_{95} = 5.3°, and the other for 12 volcanic sites is D = 350.5°, I = 25.5° with α_{95} = 7.7°, corresponding to a paleopole at 70.5°N, 292.9°E with A_{95} = 7.4°. Our new paleomagnetic results, together with reliable Cretaceous paleomagnetic data obtained from the Lhasa terrane, demonstrate that the southern margin of Asia was located at ~15.1°N during the Cretaceous. Comparison with the apparent polar wander paths (APWP) of India and the Cretaceous–Paleocene paleopoles of the Himalayan terrane suggests that the India–Asia collision was likely a complex process, which consists of the collision of the Lhasa and Himalayan terranes at 54.0 ± 2.1 Ma, the longtime subduction of an intra-oceanic basin between the Himalayan terrane and the Indian craton from ~54.0 to ~40.4 Ma, and the collision of the Himalayan terrane and the Indian craton at 40.4 ± 4.1 Ma. Comparing with the Late Cretaceous average pole of the East Asia APWP reveals that a latitudinal convergence of 780 ± 240 km has taken place between the Lhasa terrane and East Asia (the Hexi corridor) since the India–Asia collision; the amount of latitudinal shortening deduced from paleomagnetic data is very consistent with the 600–1000 km accommodated by the Cenozoic fold and thrust belts between the Lhasa terrane and Hexi corridor.

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1. Introduction

The India–Asia collision created the Himalayan–Tibetan Orogen, and thus provides a typical example for the study of continent–continent collision and associated intracontinental deformation within Asia (e.g., Yin and Harrison, 2000; Aitchison et al., 2007). The precise timing of the India–Asia collision is a key to constructing the evolution model of the Himalaya–Tibetan orogenic system (e.g., Aitchison et al., 2007). Noticeably, although many geologic and geophysical investigations have been carried out in the Himalayan–Tibetan Orogen and its neighboring regions in the last four decades (e.g., Chang and Zheng, 1973; Molnar and Tapponnier, 1975; Besse et al., 1984; Otofuji et al., 1989; Enkin et al., 1992; Meyer et al., 1998; Yin and Harrison, 2000; Tapponnier et al., 2001; Yang et al., 2002; Gilder et al., 2003; Huang et al., 2006; Wang et al., 2008; Yin, 2010; Zhu et al., 2011; Pan et al., 2012; Ran et al., 2012; Chatterjee et al., 2013; Zhu et al., 2013), the exact age of the India–Asia collision remains debated based on different methods, with estimates ranging from 70 Ma to 34 Ma (e.g., Besse et al., 1984; Rowley, 1996; Aitchison et al., 2000; Yin and Harrison, 2000; Aitchison and Davis, 2001; Ding et al., 2003; Mo et al., 2003; Ding et al., 2005; Zhu et al., 2005; Aitchison et al., 2007; Ali and Aitchison, 2008; Mo et al., 2008; Huang et al., 2010; Najman et al., 2010; Sun et al., 2010; Cai et al., 2011; Yi et al., 2011; Meng et al., 2012; Sun et al., 2012; Tang et al., 2013). Paleomagnetism can determine the plate paleolatitude as a way to constrain kinematic processes of plate movement, and thus can be used to constrain the age of the India–Asia collision by overlapping the paleolatitude of the northern margin of India with the southern margin of Asia (e.g., Chen et al., 2010; Dupont-Nivet et al., 2010; Sun et al., 2010; Yi et al., 2011; Meng et al., 2012; Tang et al., 2013). Because the Lhasa terrane was located along

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the southern margin of Asia prior to the Cenozoic India–Asia collision, Cretaceous and Paleocene–Eocene paleomagnetic data from the Lhasa terrane are essential to constrain the age of the India–Asia collision, as well as to estimate the amount of the post-collisional convergence within Asia. However, the reported paleomagnetic data from both volcanic and sedimentary rocks yield quite large uncertainties, with a significantly discrepant paleolatitudes ranging from 7°N to 32°N (e.g., Pozzi et al., 1982; Achache et al., 1984; Lin and Watts, 1988; Sun et al., 2008, 2010; Tan et al., 2010; Yi et al., 2011; Chen et al., 2012; Sun et al., 2012; Tang et al., 2013). These discrepant paleolatitudes provide the wide collision ages ranging from 65 Ma to 43 Ma, as well as the discrepant post-collisional shortening estimates ranging from 1900 km to only a few hundred kilometers. Because paleomagnetic records from volcanic rocks provide only spot readings of the geomagnetic field behavior (Tauxe, 1993), the large range of the observed paleolatitudes in volcanic rocks may reflect the secular variation of the geomagnetic field (e.g., Sun et al., 2012). Considering that compaction-induced inclination shallowing is generally present in sedimentary rocks (e.g., Gilder et al., 2001; Tan et al., 2003; Tauxe, 2005; Yan et al., 2005; Tan et al., 2007), the lower paleolatitudes observed in sedimentary rocks may thus partly be attributed to low-latitude bias due to inclination shallowing (e.g., Tan et al., 2010; Huang et al., 2013). Furthermore, the Lhasa terrane is long and narrow, with a width less than 300 km from south to north but a length more than 2000 km from east to west (Fig. 1). However, the reported Cretaceous and Paleocene–Eocene data mainly focus on the eastern part of the Lhasa terrane, with only four Cretaceous–Paleocene paleomagnetic data from its mid-western part have been published until recently (Chen et al., 1993, 2012; Meng et al., 2012; Tang et al., 2013). To better constrain the paleolatitude of the southern margin of Asia prior to the Cenozoic India–Asia collision, we carried out a combined paleomagnetic and geochronological study for the Upper Cretaceous Jingzhushan Formation (Fm) redbeds and the Lower Cretaceous Dianzhong Fm volcanic rocks in the Cuoqin area, central Lhasa terrane. Our new paleomagnetic data provide the necessary paleolatitude control on the southern margin of Asia.

2. Geological setting and sampling

The Qinghai–Tibet plateau is generally regarded as a collage of five terranes; from north to south these are the Kunlun–Qaidam, Sonpan-Ganze–Hoh Xil, Qiangtang, Lhasa and Himalayan terranes (Fig. 1). These terranes rifted from Gondwanaland and then accreted to the Asian continent (e.g., Dewey et al., 1988; Yin and Harrison, 2000; Metcalfe, 2013). The Lhasa terrane is now located between the Himalayan terrane on the south and the Qiangtang terrane on the north. Previous geological observations showed that it accreted onto the Qiangtang terrane of southern Asia during the Late Jurassic–Early Cretaceous, and was located at the southern margin of Asia before the Cenozoic India–Asia collision (e.g., Chen et al., 1993; Matte et al., 1996; Halim et al., 1998; Metcalfe, 2006; Chen et al., 2012).

The Upper Cretaceous Jingzhushan Fm, which is widely exposed in the Cuoqin sampling region (Fig. 2), is composed of purplish-red clastic sedimentary rocks intercalated with limestone on the top. Its thickness varies from several hundred meters to more than 1600 m. The Jingzhushan Fm is overlain unconformably by the Neogene Jiejunazhuo Fm redbeds, and lies over the Early Cretaceous Zenong Group (Gp) volcanic rocks. Fossils identified in this formation include *Orbitolina* (*Orbitolina* sp., *Orbitolina* (*Columnorbitolina*) cf. *alticonica* Zhang, *Orbitolina* (*Palorbitolina*) *complanata* Zhang, *Orbitolina* (*Orbitolina*) *bangoinica* Zhang) and small Foraminifera (*Cuneolina immatura* He, *Cuneolina camposaurii* Sartoni et Creascenti, *Psendocyclamina* sp., *Daxia cenomana* Cuvillier et Szakall, *Rotalipora* sp.), indicative of Late Cretaceous time (1:250,000 scale Cuoqin regional geological survey report (H45C001001), 2003). In addition to the Jingzhushan Fm redbeds, the Zenong Gp and Dianzhong Fm volcanic rocks are also widely exposed in the Cuoqin sampling area (Fig. 2). The Zenong Gp is mainly composed of gray volcanic lavas and pyroclastic rocks, with an average thickness more than 1000 m. Previous zircon U–Pb dating showed that the Zenong Gp volcanism began at ca. 130 Ma and ceased at ca. 110 Ma (Zhu et al., 2008). The Dianzhong Fm volcanic rocks are widely distributed along the southern margin of the Zenong Gp volcanic rocks, and

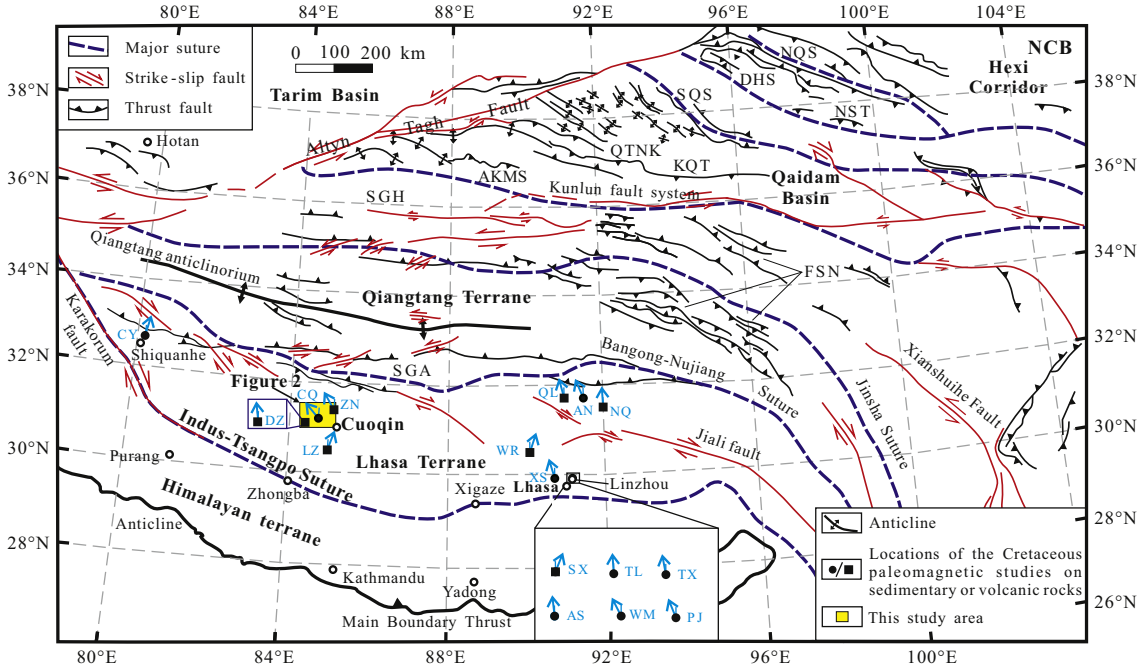


Fig. 1. Tectonic sketch map of Central Asia modified after Yin and Harrison (2000), showing sampling locations of previous Cretaceous paleomagnetic studies in the Lhasa terrane (for abbreviations see Table 3). Declination is indicated by arrows. A thin line indicates geographic north, and it is vertical because of the projection selected. Abbreviations: AKMS, Ayimaqin–Kunlun–Muztagh suture; DHS, Danghe Nan Shan suture; FSN, Fenghuo Shan–Nanqian fold and thrust belt; KQT, Kunlun–Qaidam terrane; QTNK, Qimen Tagh–North Kunlun thrust system; NCB, North China block; NQS, North Qilian Suture; NST, Nan Shan thrust belt; SGA, Shiquanhe–Gaize–Ando thrust system; SGH, Songpan-Ganzi–Hoh Xil terrane; SQS, South Qilian suture; NCB, North China Block.

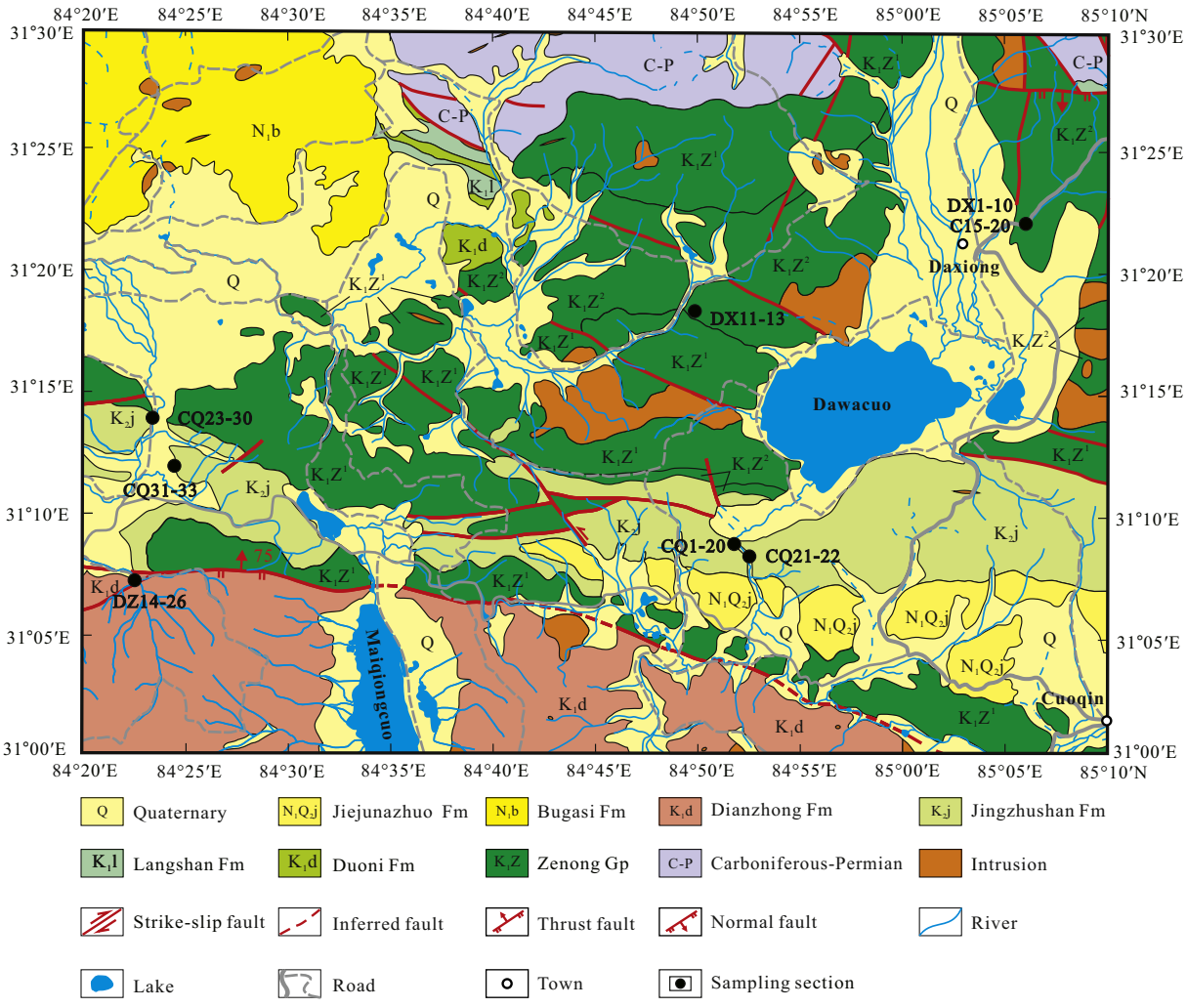


Fig. 2. Simplified geological map of the sampling area.

also mainly consist of green–gray volcanic lavas and pyroclastic rocks. Noticeably, although the Dianzhong Fm volcanic rocks in the Linzhou basin has been widely accepted to have a Paleocene age (e.g., Zhou et al., 2004; Lee et al., 2007; Liang et al., 2010), our new U–Pb zircon ages ranging from ca. 121 to 117 Ma (see Section 3), as well as new ^{40}Ar–^{39}Ar ages ranging from ca. 99 to 93 Ma (Tang et al., 2013), show that the Dianzhong Fm volcanic rocks in the Cuoqin area were formed during the Cretaceous. The earliest folding of the Lower and Upper Cretaceous strata occurred in the latest Early Cretaceous and latest Paleocene, respectively (Tao et al., 2008).

A total of 493 oriented cores distributed at 46 paleomagnetic sites were collected from five localities, located ~20–60 km northwest of the county of Cuoqin (Fig. 2). 356 cores from 33 sites (CQ1–CQ33) distributed at four localities were collected from the Jingzhushan Fm purplish-red siltstones. 137 cores from 13 sites (DZ14–DZ26) were drilled from the Dianzhong Fm volcanic rocks. These 13 volcanic sites are stratigraphically independent, and were collected from a continuously exposed natural outcrop along a small river. Bedding attitudes for both the Dianzhong Fm volcanic rocks and the Jingzhushan Fm redbeds are quite obvious at all the sampling localities (Fig. 3). The bedding attitudes and coordinates of each site are presented in Tables 1 and 2.

Generally, each paleomagnetic site consists of at least 10 independent core samples and spans several meters in stratigraphic thickness — or at least one lava flow (or several tuff layers) (Fig. 3). All the core samples were collected using a portable gasoline-powered drill, and they were oriented using a magnetic compass. For the cores of the Dianzhong Fm volcanic rocks, a sun compass was also used to measure their declination. Comparing these two oriented results revealed a declination difference less than 2°, indicating that the local magnetic disturbance can be neglected.

In addition, two fresh block samples located at the uppermost (DZ14) and lowermost (DZ26) paleomagnetic sites of the sampling Dianzhong Fm volcanic rocks were collected for zircon U–Pb chronology.

3. U–Pb zircon dating

Zircon grains were separated at the Langfang Laboratory of Geophysical Exploration, Geological Exploration Bureau of Hebei Province, Ministry of Land and Mineral Resources. The U–Pb zircon dating was carried out using LA-ICPMS at the Tianjin Institute of Geology and Mineral Resources. Analytical procedures for zircons were similar to those documented by Yuan et al. (2003). The analytical spots of zircons were chosen by photographic inspection at positions that show clear oscillatory zoning and are free of inclusions and cracks. U–Th–Pb concentrations were calibrated using NIST612 as an external standard. Fractionation correction of the U–Pb isotope was conducted using GJ-1 as an external standard (Jackson et al., 2004). Common lead was corrected according to the method described by (Andersen, 2002).

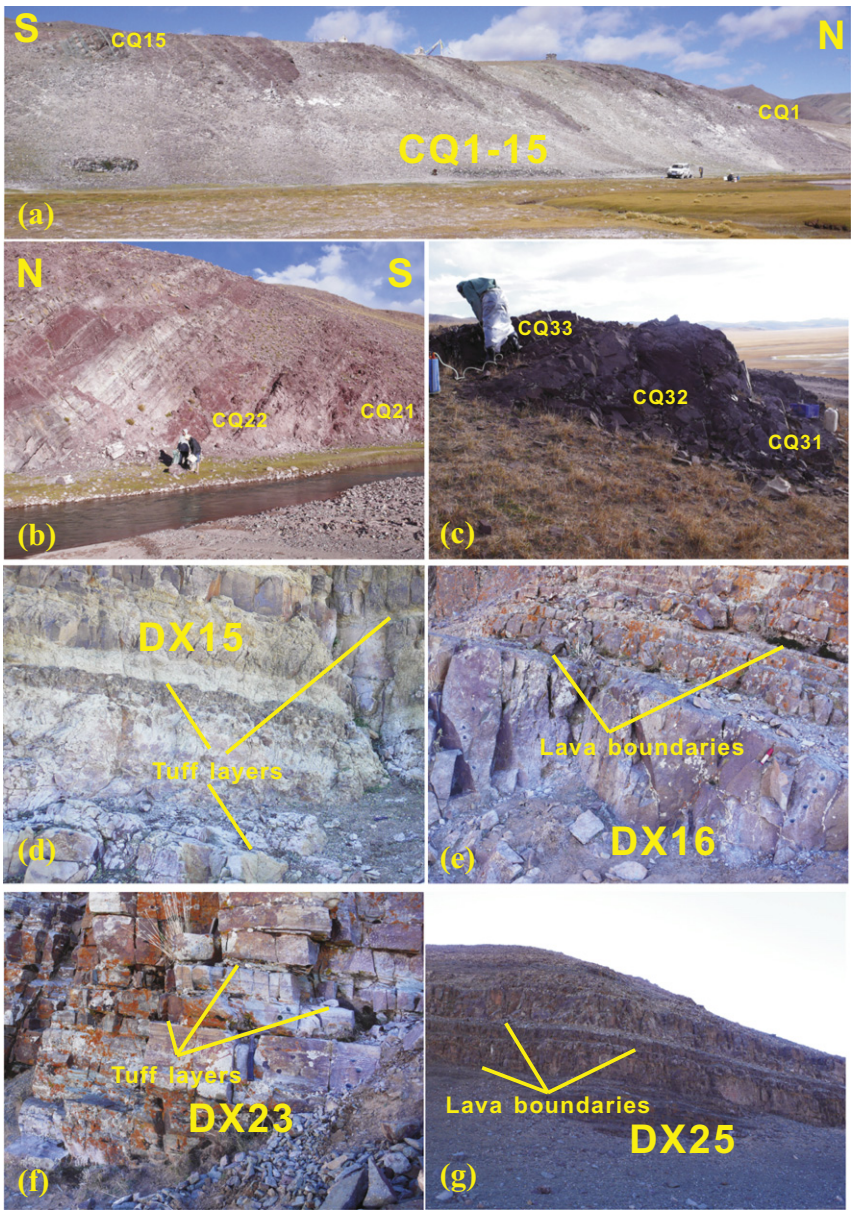


Fig. 3. Photographs in (a–c) and (d–f) show the field outcrops of the Upper Cretaceous Jingzhushan Fm redbeds and the Lower Cretaceous Dianzhong Fm volcanic rocks, respectively.

Data processing was carried out using the ICPMS DataCal (Liu et al., 2010) and Isoplot (Ludwig, 2003) programs.

All the separated zircon grains are euhedral to subhedral, with prismatic shapes at an aspect ratio of ~2 (Fig. 4). These, together with an exhibition of clear oscillatory zoning, indicate that the zircon grains are igneous in origin. The results of the U–Pb zircon analyses from the samples DZ14 and DZ26 are presented in Fig. 5. These two samples, which were collected from the uppermost (DZ14) and lowermost (DZ26) paleomagnetic sites of our sampling Dianzhong Fm volcanic rock sequence, yield concordant ages of 117.03 ± 0.51 Ma and 121.01 ± 0.66 Ma, revealing that the stratigraphic sequence of the sampling Dianzhong Fm volcanic rocks have not been overturned during the Cenozoic orogeny. Furthermore, our new zircon ages confirm that the sampled Dianzhong Fm volcanic sequences were erupted during the Early Cretaceous, not during the Paleogene. Similar U–Pb zircon ages, as well as similar lithologic characters (1:250,000 scale Cuoqin regional geological survey report (H45C001001), 2003) obtained from the Dianzhong Fm and adjacent Zenong Gp volcanic rocks suggest that they might be the same rock sequence erupted during the Early Cretaceous.

4. Paleomagnetic analyses and results

Paleomagnetic oriented cores were cut into 2.2-cm-long specimens in the laboratory. Remanent magnetization was measured using a 2G-755-4K cryogenic magnetometer or a JR-6A spinner magnetometer at the Paleomagnetic and Environmental Magnetism Laboratory of the China University of Geosciences, Beijing. Paleomagnetic specimens underwent a stepwise thermal demagnetization up to 580 °C for the Dianzhong Fm volcanic rocks or 680 °C for the Jingzhushan Fm redbeds in an ASC-TD 48 furnace with an internal residual field less than 10 nT. Demagnetization temperature intervals are generally 100–50 °C in the low temperature part, and shift to 20–5 °C at higher temperatures. All remanence measurement and thermal demagnetization were performed within a magnetic shielded room with a residual field less than 300 nT. In order to identify the magnetic minerals and ensure the

Table 1
Paleomagnetic results from the Upper Cretaceous Jingzhushan Formation redbeds.

Site	Slat (°N)	Slon (°E)	Strike/dip (°)	n/N	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95} (°)	Plat (°)	Plon (°)
CQ1	31.15	84.86	277.5/64	11/11	275.3	34	308.1	15.9	812.7	1.6	36.6	340.8
CQ2	31.15	84.86	277.5/64	10/10	270.8	36.5	309.2	20.1	294.5	2.8	38.7	342.5
CQ3	31.15	84.86	277/66	12/12	267.1	40.8	313.1	22.6	173.4	3.3	42.7	341.4
CQ4	31.15	84.86	277/66	9/10	260.7	39.4	310.6	27.2	76.8	5.9	41.9	346.1
CQ5	31.15	84.86	277/66	10/10	262.4	37.2	308.5	25.4	129.6	4.3	39.6	346.2
CQ6	31.15	84.86	273/65	8/9	260.8	33.5	300.6	23.2	90.1	5.9	32.3	349.6
CQ7	31.15	84.86	278/64	9/9	271.3	34.9	308.1	19.7	119.3	4.7	37.7	343
CQ8	31.15	84.86	278/63	8/9	274.8	37.1	311	18.3	104.8	5.4	39.7	340.2
CQ9	31.15	84.86	278/61	9/12	267.4	42.8	314.2	26.6	406.1	2.6	44.8	343.3
CQ10	31.15	84.86	278/59	10/10	265.8	36.4	306.1	26.9	221	3.3	38	348.7
CQ11	31.15	84.86	278/63	10/12	264.3	40.2	311.6	27	248.8	3.1	42.7	345.3
CQ12	31.15	84.86	280/62	8/10	269.3	35	308.4	23.8	119.2	5.1	39.1	345.3
CQ13	31.15	84.86	277/60	9/9	265	25.3	294.3	22.1	144	4.3	26.6	352.4
CQ14	31.15	84.86	273/63	9/9	262.6	33	299.8	22.5	147.3	4.3	31.4	349.6
CQ15	31.15	84.86	273/63	8/10	263.9	34.5	301.7	21.9	487.3	2.5	32.9	348.2
CQ16	31.15	84.86	279/62	9/9	261.9	38.1	309.1	29.6	111.3	4.9	41.3	348.7
CQ17	31.15	84.86	279/62	10/10	267.6	35.1	307.3	24.3	134.9	4.2	38.3	346.3
CQ18	31.15	84.86	274/62.5	8/10	271	32.1	302	16.5	165.7	4.3	31.6	345
CQ19	31.15	84.86	276/61	10/11	263.8	33.3	301.7	24.9	144.8	4	33.7	349.9
CQ20	31.15	84.86	274/65	7/9	258.9	29.1	296.2	24.3	131.4	5.3	28.8	352.6
CQ21	31.14	84.87	277/51	10/11	295.9	63.6	338.8	26.8	109.1	4.6	64.2	318.5
CQ22	31.14	84.87	267.5/44	10/10	300.4	52.5	324.6	20	64.5	6.1	51.2	330.2
CQ23	31.23	84.39	113/48	8/8	348.5	−2.7	339.1	35.5	68.5	6.7	67.9	327.7
CQ24	31.23	84.39	115/50	9/9	354.8	5.8	338.5	46.4	63.8	6.1	71	349.2
CQ25	31.23	84.39	112/53	9/9	347.3	6.6	327.3	46.2	30.4	9.5	61.4	355.4
CQ26	31.23	84.39	116/54	9/9	353.8	9	330.7	50.2	73.7	6	65	1.5
CQ27	31.23	84.39	123/52	7/8	352.9	8.1	332.1	43.1	54.3	8.3	64.7	347.3
CQ28	31.23	84.39	121/53	6/6	349	5.3	330.2	40.3	108.5	6.5	62.3	344.4
CQ29	31.23	84.39	119/53	8/8	356.2	4	338.7	45.4	91.5	5.8	70.9	346.6
CQ30	31.23	84.39	118/55	8/9	354.8	0.9	338.4	44	184.3	4.1	70.3	343.7
CQ31	31.20	84.41	43/33	7/9	340.9	15.6	351.4	43.6	21.5	13.3	80.5	319.4
CQ32	31.20	84.41	43/33	7/9	342.8	3.9	348.8	32	42.5	9.4	72.8	303.3
CQ33	31.20	84.41	39/36	9/9	335.1	−0.8	340	31.1	93.1	5.4	66.8	320.6
Site-mean (N = 33 sites)				291/315	297.7	33.2	316.8	30.2	22.4	5.4	49.0	344.3
											K = 22.8	A_{95} = 5.3

Notes: Slat and Slon, latitude and longitude of sampling site; Plat and Plon, latitude and longitude of pole; n/N, number of samples used to calculate mean and measured; Dg, Ig, Ds, Is, declination and inclination in geographic and stratigraphic coordinates, respectively; k (K), the best estimate of the precision parameter; a_{95} (A_{95}), the radius that the mean direction (pole) lies within 95% confidence.
The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: ks/kg = 4.95 > F(2 * (n2 − 1), (n1 − 1)) at 5% and 1% point = 1.51 and 1.80, respectively. The McFadden (1990) fold test is also positive at 95% and 99% confidence levels at "Xi1" test: critical Xi at 95% and 99% = 6.68 and = 9.45, respectively. "Xi1" IS = 35.77, "Xi1" TC = 2.64.

reliability of paleomagnetic data, isothermal remanent magnetization (IRM) and back-field demagnetization of saturation IRM (SIRM) were acquired using an IM10-30 pulse magnetizer, and was measured using a JR-6A spinner magnetometer. Thermomagnetic analyses from the same specimens were performed using a variable field translation balance at the Paleomagnetism and Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences. High-temperature component (HTC) directions of all the specimens were determined using principal component analysis (Kirschvink, 1980), and site-mean directions were calculated using the Fisherian statistics (Fisher, 1953). Paleomagnetic data were analyzed using the computer program packages developed by Enkin (1990) and Cogné (2003).

4.1. Upper Cretaceous Jingzhushan Fm redbeds

IRM acquisition curves of the Jingzhushan Fm redbeds present a slow increase, and the saturation is not far reached even at 2.5 T (Fig. 6a and b), revealing that high-coercivity magnetic carriers are dominant. Progressive removal of the SIRM by applying reverse fields indicates that the maximum coercive force is ca. 500–600 mT. Thermomagnetic curves of the same specimen displays the clear Curie temperatures of high-coercivity hematite at 680 °C (Fig. 7a and b).

Representative demagnetization diagrams of the Jingzhushan Fm redbeds are shown in Fig. 8a–p. Some specimens can have a low-temperature component (LTC) between NRM and 300 °C. This LTC is close to the present geomagnetic field direction, and thus should be a viscous remanent magnetization of the present geomagnetic field direction. After removing this LTC, a HTC decay toward the origin can be isolated between 400 °C and 670–680 °C. Significantly, a specimen from the CQ27 shows a reverse polarity HTC that is almost antipodal to other normal polarity HTCs (Fig. 8o). A high unblocking temperature up to 670–680 °C, together with the above-mentioned rock-magnetic results, shows that the HTC is carried by hematite. Each site-mean direction of the HTCs is listed in Table 1. The site-mean direction of 33 sites is Dg = 297.7°, Ig = 33.2°, kg = 4.5, and α_{95} = 13.3° in situ and Ds = 316.8°, Is = 30.2°, ks = 22.4, and α_{95} = 5.4° after tilt-correction (Fig. 9a and Table 1), corresponding to a paleopole at 49.0°N, 344.3°E with A_{95} = 5.3°. It passes both the McElhinny (1964) and McFadden (1990) fold tests at 95 and 99% confidence levels, yielding a maximum grouping at 96% unfolding (Fig. 10a). This, combined with dual polarity HTC observed from the site CQ27, indicates that the HTC should be a prefolding primary magnetization before the latest Paleocene.

Obviously, the declinations of the site-mean directions from sites CQ1–20 indicate a difference of ca. 30° relative to those from the sites CQ21–33 (Table 1 and Fig. 9), which likely resulted from the local vertical-axis rotation of the sampling locations associated with the widely distributed faults (Fig. 2). After a 30° clockwise rotation of the site-mean directions for sites CQ1–20, the tilt-corrected Fisher precision parameter (k) increases clearly from 22.4 to 51.0 (Fig. 9b). Following the procedure of Arason and Levi (2010), the inclination-only (and corresponding to paleolatitude-only) mean of 33 site-mean directions is 29.8 ± 4.4°, which is very consistent with the inclination of 30.2 ±

Table 2
Paleomagnetic results from the Lower Cretaceous (117–121 Ma) volcanic rocks.

Site ID	Slat (°N)	Slon (°E)	Strike/dip (°)	n/N	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95} (°)	Plat (°)	Plon (°)
This study												
DZ14	31.12	84.37	224/28	10/10	24.3	31.8	12.0	19.3	61.0	6.2	66.1	234.1
DZ15	31.12	84.37	225/24	11/11	19.3	30.1	10.0	17.8	29.9	8.8	66.1	239.3
DZ16[a]	31.12	84.37	218/21.5	10/10	116.7	28.8	112.6	49.7	93.3	5.0	−1.2	137.1
DZ17	31.12	84.38	218/21.5	9/10	32.4	49.6	10.4	43.3	100.0	5.2	79.1	204.6
DZ18	31.12	84.38	218/21.5	10/11	10.6	36.1	360.0	24.4	238.6	3.1	71.7	264.4
DZ19	31.12	84.38	218/21.5	10/11	347.2	36.4	340.5	18.8	202.7	3.4	61.9	308.7
DZ20	31.12	84.38	218/21.5	10/12	349.5	34.3	343.0	17.3	92.7	5.0	62.7	303.4
DZ21	31.12	84.38	218/21.5	11/11	348.3	36.4	341.5	19.1	31.5	8.3	62.6	307.2
DZ22	31.12	84.38	202/6	9/10	342.9	31.0	340.4	27.1	136.2	4.4	65.4	315.8
DZ23	31.12	84.38	225.5/12	8/9	346.0	33.6	342.9	23	123.6	5.0	65.2	307.7
DZ24	31.12	84.38	227.5/10.5	8/9	347.7	42.0	343.9	32.8	104.5	5.4	70.3	315.9
DZ25	31.12	84.38	232/12	10/10	350.2	41.4	346.3	30.6	76.8	5.5	70.8	307.9
DZ26	31.12	84.38	224/26	10/11	348.1	49.7	338.0	26.8	105.1	4.7	63.7	319.3
Site-mean 1 (N = 12 sites)				116/125	358.5	38.9	350.5	25.5	32.8	7.7	70.5	292.9
											K = 35.1	A_{95} = 7.4
Previous study by Chen et al. (2012)												
C15	31.37	85.1	0/48	9/9	319.2	−2.3	328.3	27.3	29.8	9.6	56.5	332.1
C16	31.37	85.1	6/55	6/6	321.5	−4.2	333.1	32.0	36.9	11.2	61.9	331.5
C17	31.37	85.1	6/55	7/8	323.7	−6.3	333.2	29.0	46.9	8.9	60.9	328.3
C18	31.37	85.1	22/49.5	10/11	337.0	0.3	347.7	25.9	21.0	10.8	69.0	300.4
C19	31.37	85.1	20/54	7/9	315.6	−8.0	323.9	39.9	24.2	12.5	56.9	348.9
C20	31.37	85.1	5/48	11/11	133.7	−0.2	145.3	−35.5	26.0	9.1	−56.7	162.9
DX1	31.37	85.1	20/55	9/10	312.0	−13.6	317.2	37.1	148.1	4.2	50.4	350.1
DX2	31.37	85.1	20/55	11/11	318.5	−18.7	321.4	29.8	71.0	5.5	51.7	340.5
DX3	31.37	85.1	20/55	7/7	307.9	−13.4	312.6	38.7	44.5	9.1	46.9	354.2
DX4	31.37	85.1	16/56	11/11	306.3	−4.3	316.6	47.2	119.3	4.2	52.5	2.0
DX5	31.37	85.1	14.5/55	8/9	326.7	−1.4	341.0	36.3	89.7	5.9	69.6	326.4
DX6	31.37	85.1	14.5/55	10/10	323.7	−4.3	335.8	36.1	38.0	7.9	65.5	333.4
DX7	31.37	85.1	20/53	10/12	315.5	−7.2	323.8	39.7	310.0	2.7	56.7	348.8
DX8	31.37	85.1	20/53	10/10	150.1	2.2	162.7	−35.9	40.7	7.7	−70.7	142.8
DX9	31.37	85.1	16/54	10/10	316.0	−8.0	324.7	37.6	145.2	4.0	56.9	345.7
DX10	31.37	85.1	10/53	12/12	314.8	−12.7	320.2	30.4	115.4	4.1	50.9	342.0
DX11	31.31	84.8	271/4	9/10	305.4	32.6	307.3	30.2	51.1	7.3	40	350.0
[a]DX12	31.31	84.8	270/4	9/9	94.4	2.2	94.3	2.5	67.9	8.2	−3.0	171.5
DX13	31.31	84.8	3/5	5/10	325.5	41.8	329.3	44.7	145	6.4	62.7	352.1
Site-mean 2 (N = 18 sites)				162/178	318.5	−2.1	327.0	35.7	59.3	4.5	58.2	341.9
											K = 35.1	A_{95} = 4.6
Combined site-mean (N = 30 sites)				277/303	332.0	15.0	336.9	32.2	25.0	5.4	64.9	328.0
											K = 24.0	A_{95} = 5.5

Notes: Slat and Slon, latitude and longitude of sampling site; Plat and Plon, latitude and longitude of pole; n/N, number of samples used to calculate mean and measured; Dg, Ig, Ds, Is, declination and inclination in geographic and stratigraphic coordinates, respectively; k (K), the best estimate of the precision parameter; a_{95} (A_{95}), the radius that the mean direction (pole) lies within 95% confidence.
Fold tests for the combined site-mean direction of 30 sites are positive: (1) The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: ks/kg = 3.67 > F(2 * (n2 − 1), (n1 − 1)) at 5% and 1% point = 1.55 and 1.86, respectively; (2) The McFadden (1990) fold test is also positive at 95% and 99% confidence levels at "Xi1" test: critical Xi at 95% and 99% = 6.37 and 9.01, respectively. "Xi1" IS = 21.08, "Xi1" TC = 3.18.
[a] Sites were not used to calculate the final mean direction.

5.4° of the Fisher site-mean direction. Therefore, the site-mean pole (49.0°N, 344.3°E with A_{95} = 5.3°) obtained from 33 Late Cretaceous redbed sites can be used to constrain the pre-collisional paleolatitude of the Lhasa terrane.

4.2. Lower Cretaceous Dianzhong Fm volcanic rocks

IRM acquisition curves of the Dianzhong Fm volcanic rocks show a rapid increase, and more than 90% of the saturation IRM is acquired below 200 mT (Fig. 6c and d). This, combined with the maximum coercive force less than 40 mT, indicates that low-coercivity magnetic carriers are dominant in the studied specimen. Thermomagnetic curves further reveal the Curie temperature of low-coercivity magnetite at 580 °C (Fig. 7c and d).

A LTC between NRM and 200–300 °C and a HTC between 500–530 °C and 570–580 °C (Fig. 8q–u) can be determined from most specimens from the Dianzhong Fm volcanic rocks, which together with the above-mentioned rock-magnetic results shows that the low-coercivity magnetite is a major magnetic carrier for the HTC in the Dianzhong Fm volcanic rocks. Remarkably, a specimen from site DZ14 reveals a normal polarity LTC that is consistent with other specimens from site DZ14, but it shows a reversed polarity HTC that is almost antipodal to the normal polarity HTCs isolated from other specimens of site 14 (Fig. 8r and s). Table 2 summarizes each site-mean direction of the HTCs. Site DZ16, whose site-mean direction is proved to be an outlier according to the Jackknife test (Cogné, 2003), was excluded from the final site-mean. The site-mean direction of 12 sites is Dg = 358.5°, Ig = 38.9°, kg = 28.9, and α_{95} = 8.3° in situ and Ds = 350.5°, Is = 25.5°, ks = 32.8, and α_{95} = 7.7° after tilt-correction (Fig. 11 and Table 2). Due to the 12 paleomagnetic sites sampled from a monoclinal sequence, the fold tests are therefore inapplicable. However, the Fisher precision parameter (k) increases from 28.9 to 32.8 after tilt-correction, which, together with dual polarity HTC observed from site DZ14, suggests that the HTC obtained from the Dianzhong Fm volcanic rocks should be a prefolding primary magnetization before the latest Early Cretaceous.

Significantly, Chen et al. (2012) reported reliable Early Cretaceous paleomagnetic results from the Zenong Gp volcanic rocks in the Cuoqin area. These paleomagnetic data from a similar age range and adjacent areas provide a good chance for performing field tests to assess the reliability of the Dianzhong Fm volcanic data. The combined site-mean direction of 30 sites is Dg = 332.1°, Ig = 15.0°, kg = 6.8, and α_{95} =

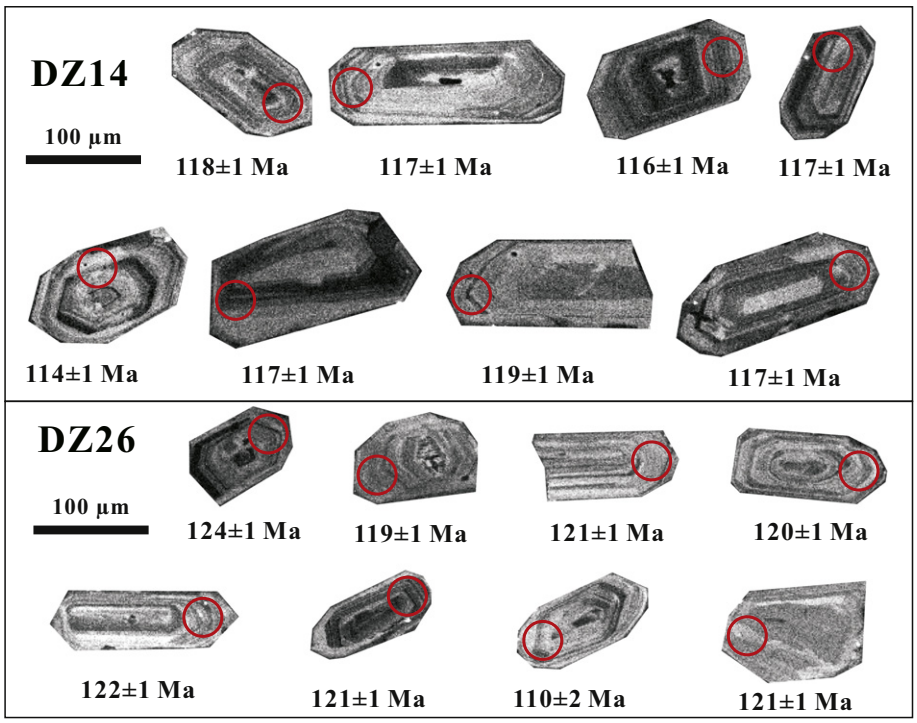


Fig. 4. Cathodoluminescence images of representative zircons for the DZ14 and DZ26 sites at shown positions and $^{206}Pb/^{238}U$ ages of the individual analyzed spots.

10.9° in situ and Ds = 336.9°, Is = 32.2°, ks = 25.0, and α_{95} = 5.4° after tilt-correction, corresponding to a paleopole at 64.9°N, 328.0°E with A_{95} = 5.5° (Table 2). This combined site-mean direction passes both the McElhinny (1964) and McFadden (1990) fold tests at 95% and 99% confidence levels (Table 2) and yields a maximum grouping at 100% unfolding (Fig. 10b), further supporting that the HTC obtained from the Early Cretaceous Dianzhong Fm volcanic rocks was a prefolding primary magnetization before the latest Early Cretaceous.

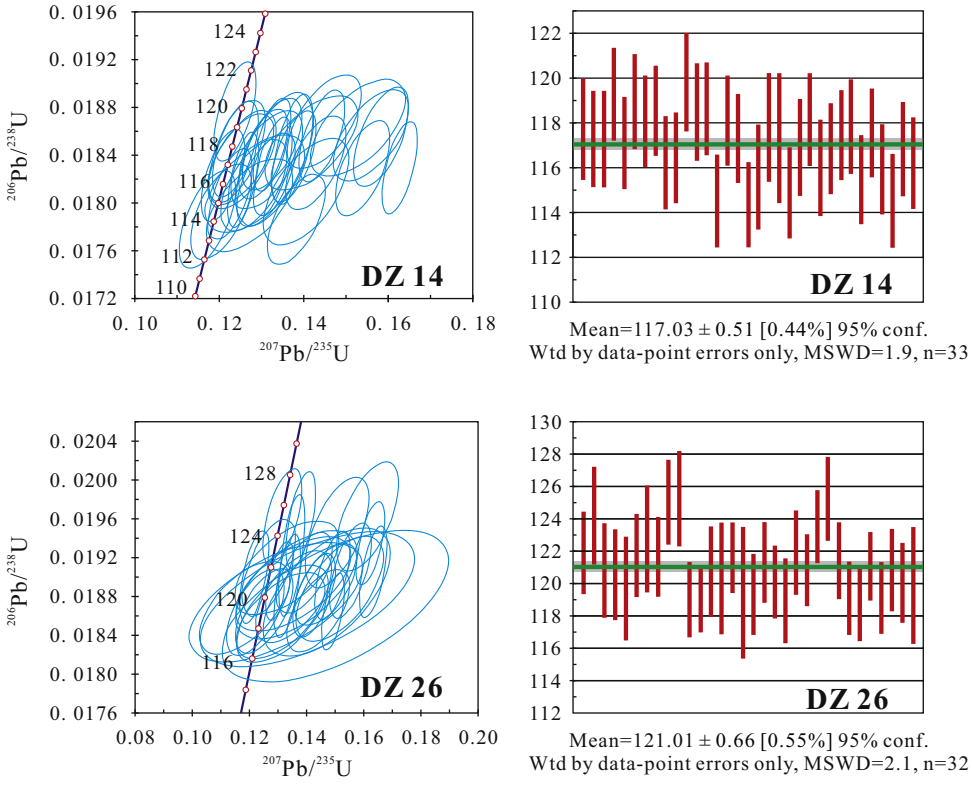


Fig. 5. Concordia plots of U–Pb zircon geochronology results for the DZ14 and DZ26 sites. Green bars in right panels show weighted mean $^{206}Pb/^{238}U$ ages and red columns represent single zircon analyses.

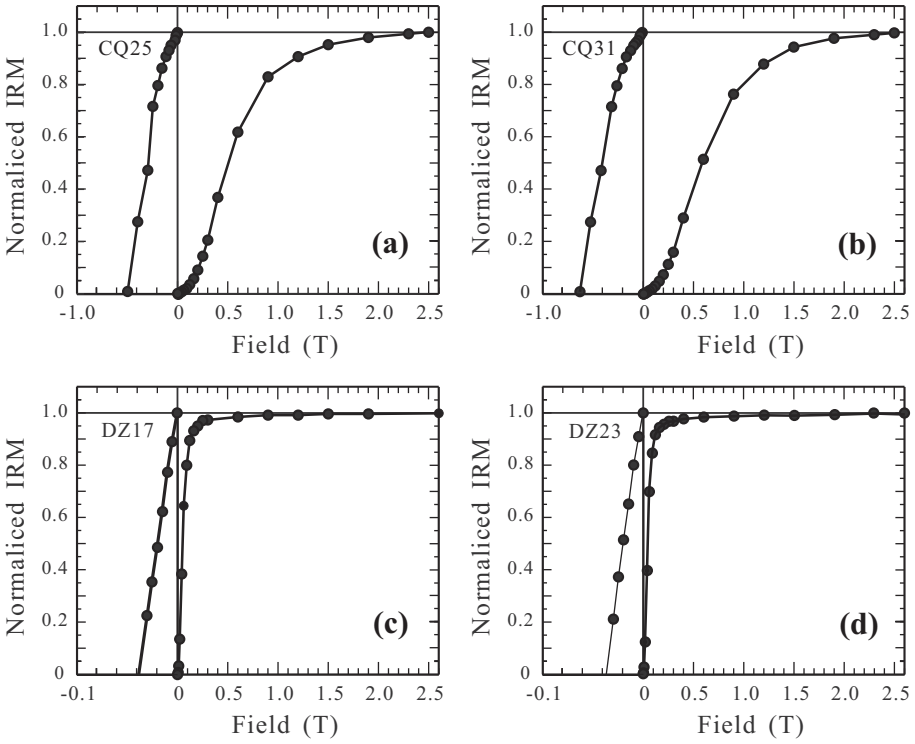


Fig. 6. Isothermal remanent magnetization (IRM) acquisition curves and back-field demagnetization of saturation IRM (SIRM) for the Jingzhushan Fm redbed (a, b) and Dianzhong Fm volcanic (c, d) specimens.

Because paleomagnetic records from volcanic rocks provide only spot readings of geomagnetic field behavior (Tauxe, 1993), we need test whether they have averaged out paleosecular variation. A widely accepted approach is to compare the virtual geomagnetic pole (VGP) scatter with those expected by McFadden et al. (1991). The sampling Dianzhong Fm volcanic rocks yield two U–Pb zircon ages of 117.03 ± 0.51 Ma and 121.01 ± 0.66 Ma which fall into the Cretaceous Normal Polarity Superchron (Ogg, 2012). The HTC directions of 12 volcanic paleomagnetic sites yield a VGP scatter of 13.7 which is higher than the expected VGP scatter of 8.7 for 80–110 Ma, but clearly lower than 18.5 for 110–195 Ma. This, together with the sampling volcanic rocks spanning a long time from 121.01 ± 0.66 Ma to 117.03 ± 0.51 Ma as determined by the U–Pb zircon dating (Fig. 5), suggests that new paleomagnetic data obtained from the Early Cretaceous Dianzhong Fm volcanic rocks may average out paleosecular variation.

5. Discussion

5.1. Position of the southern margin of Asia before the India–Asia collision

Table 3 and Fig. 12 summarize available Cretaceous paleomagnetic results from the Lhasa terrane, and their paleolatitudes have been calculated for a reference point in Zhongba county (29.77°N, 84.03°E) where it is just located in the present-day Indus–Tsangpo suture. Seven paleomagnetic results obtained from volcanic rocks provide paleolatitudes of 19.7° ± 11.1°N and 18.3° ± 4.0°N (Lin and Watts, 1988) for the Qelico (QL) and Nagqu (NQ) volcanic rocks in the northeastern Lhasa terrane, 11.8° ± 5.1°N (Sun et al., 2008) and 21.7° ± 3.3°N (Tan et al., 2010) for the Woronggou (WR) and Shexing (SX) volcanic rocks in the southeastern Lhasa terrane, and 19.0° ± 4.6°N (Chen et al., 2012), 8.0° ± 5.1°N (Tang et al., 2013), and 12.4° ± 7.4°N (this study) for the Zenong (ZN), Linzizong Gp (LZ) and Dianzhong Fm (DZ) volcanic rocks in the central Lhasa terrane. Because paleomagnetic records from volcanic rocks provide only spot readings of the geomagnetic field behavior (Tauxe, 1993), the large range of the observed paleolatitudes from the Cretaceous volcanic rocks may reflect the secular variation of the geomagnetic field (e.g., Sun et al., 2012). Considering that an accurate paleolatitude, especially one obtained from volcanic rocks, must depend on sufficient paleomagnetic sites to fully average out paleosecular variation of the geomagnetic field (Van der Voo, 1990; Tauxe, 1993), as well as that local vertical-axis rotations have occurred within the Lhasa terrane as suggested by the Cretaceous poles along a small circle centered on the reference point and by our new paleomagnetic data (Figs. 1 and 12), we first obtain the inclination-only (and corresponding to paleolatitude-only) mean following the method of Arason and Levi (2010) and then calculate the corresponding paleolatitude. Except for 9 Nagqu sites whose site-mean directions are very doubtable due to a remagnetization or an imprecise attitude determination of the individual sites as pointed out by Lin and Watts (1988), the 45 Early Cretaceous (K1) and 42 Late Cretaceous (K2) volcanic paleomagnetic sites observed in the Lhasa terrane yield the inclination-only means of 27.6° ± 4.0° and 29.2° ± 5.9°, respectively, corresponding to the paleolatitudes of 14.6° ± 2.4° and 15.6° ± 3.6° (Supplementary material, Table S1). These two paleolatitudes come from a large number of lava flows and/or tuff layers that yield dual polarity HTCs (Table 3), and thus should be two reliable paleolatitude estimates for the southern margin of Asia during the Early and Late Cretaceous volcanism. The same paleolatitude was observed from 45 reliable Early Cretaceous (K1) and 42 reliable Late Cretaceous (K2) volcanic paleomagnetic sites (Table 3), indicating that the southern margin of Asia remained a relatively stable latitude during the whole Cretaceous.

Although some Cretaceous paleomagnetic investigations on basalt flows intercalated with redbeds showed that the two site-mean directions for redbeds and basalt flows are quite consistent (Sun et al., 2006; Li et al., 2013), the possible inclination shallowing is still a critical problem for the validity of paleomagnetic data from redbeds (e.g., Tan et al., 2010). To test whether new paleomagnetic data from the Jingzhushan Fm redbeds have suffered from the compaction-induced inclination shallowing, we compare the inclination-only mean of site-mean directions and its corresponding paleolatitude for redbed sites

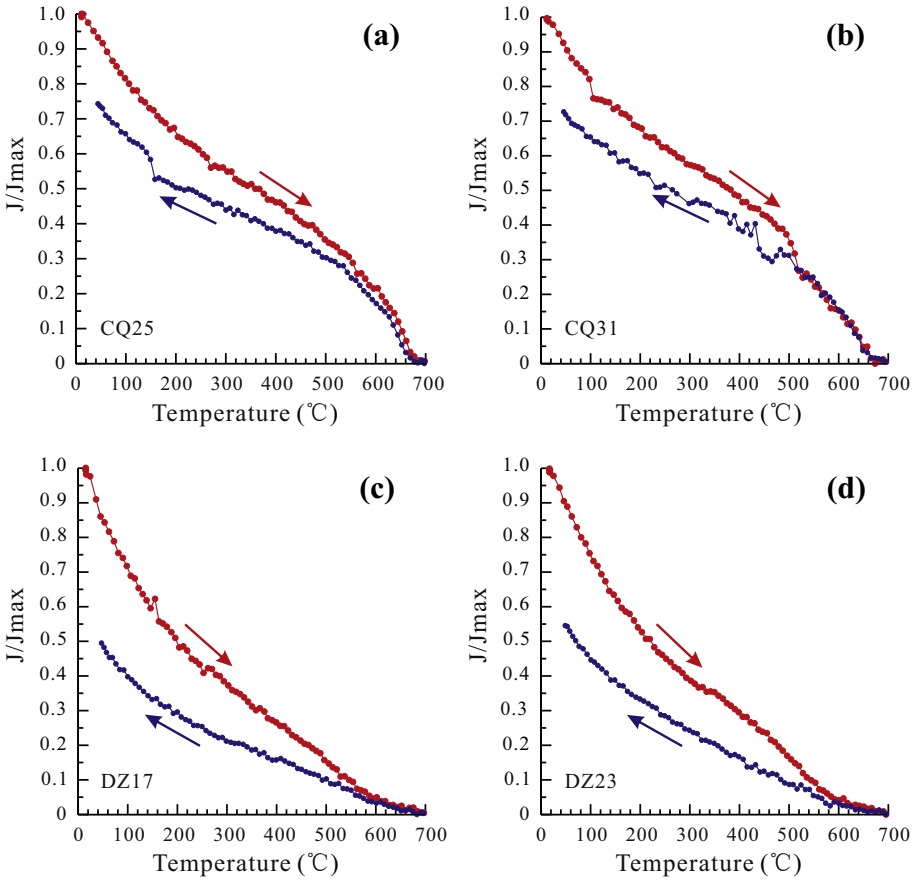


Fig. 7. Thermomagnetic curves of the Jingzhushan Fm redbed (a, b) and Dianzhong Fm volcanic (c, d) specimens.

with those for reliable volcanic sites. Our 33 Jingzhushan redbed sites yield an inclination-only mean of 28.8° ± 4.2° and a corresponding paleolatitude of 15.4° ± 2.5°N for the reference point (29.77°N, 84.03°E), which are significantly higher than 16.0° ± 7.2° and 8.2° ± 3.8°N deduced from the 14 Late Cretaceous volcanic sites in the Cuoqin sampling area, but very consistent with 29.2° ± 5.9° and 15.6° ± 3.6°N for all the 42 Late Cretaceous volcanic sites in the Lhasa Terrane (Supplementary material, Table S1). Therefore, these new paleomagnetic data obtained from the Jingzhushan Fm redbeds do not suffer from the compaction-induced inclination shallowing. In addition to our new Jingzhushan Fm data, there are 7 redbed paleomagnetic data including 89 paleomagnetic sites that have been obtained from the Upper Cretaceous Shexing Fm, which has been previously referred as the Takena Fm, in the eastern Lhasa terrane. In fact, all the 8 Late Cretaceous redbed poles yield a paleolatitude range from 8.4° ± 3.5°N to 24.0° ± 5.6°N that is very similar with 8.0° ± 5.1°N to 21.7° ± 3.3°N from 7 Cretaceous volcanic poles (Table 3), and all the 122 Late Cretaceous redbed sites provide an inclination-only mean of 28.0° ± 2.1° and a corresponding paleolatitude of 14.9° ± 1.3°N that are also consistent with 29.2° ± 5.9° and 15.6° ± 3.6°N for all the 42 Late Cretaceous volcanic sites in the Lhasa Terrane (Supplementary Material, Table S1), suggesting that the compaction-induced inclination shallowing is insignificant in the Upper Cretaceous redbeds of the Lhasa terrane.

Furthermore, Chen et al. (1993) reported a Cretaceous limestone paleopole from three sites (22 specimens) in the western Lhasa terrane. This limestone paleopole yields a paleolatitude of 10.0° ± 13.1°N. Although it is also comparable with those from redbeds within the 95% confidence level, we do not use this limestone paleopole for further discussion considering that it is from only 22 specimens, with a large error bar and without a robust field test.

Table 3 also lists 9 available Paleocene–Eocene paleomagnetic results from the Lhasa terrane. Obviously, these 9 paleomagnetic data also yield scatter paleolatitudes from 7.1° ± 8.5°N to 32.3° ± 4.8°N (Table 3). The large range of the paleolatitudes from the Linzizong volcanic rocks may reflect the secular variation of the geomagnetic field (e.g., Sun et al., 2012), and the lower paleolatitudes observed in sedimentary rocks can be attributed to low-latitude bias due to inclination shallowing (e.g., Tan et al., 2010; Huang et al., 2013). Furthermore, because the sampling Linzizong Gp volcanic rocks span a long time range from 64 to 40 Ma, the large range of the paleolatitudes from the Linzizong volcanic rocks, such as those from the Panna Fm volcanic rocks that erupted after the India–Asia collision age, may be partly attributed to the polar wander that occurred during the Linzizong Gp volcanism (e.g., Yi et al., 2011).

5.2. The India–Asia collision process

The Himalayan terrane is generally accepted to be the extent of Greater India (e.g., Ali and Aitchison, 2005; Yi et al., 2011; Meng et al., 2012), but some studies suggested that an intra-oceanic basin existed between the Himalayan terrane and the Indian craton during the Cretaceous and Paleocene (e.g., Ali and Aitchison, 2008; van Hinsbergen et al., 2012; Metcalfe, 2013). Considering that the widely accepted geological boundary between India and Asia is located in the present-day Indus–Tsangpo suture zone that separates the Himalayan terrane on the south from the Lhasa terrane on the north (e.g., Yin and Harrison, 2000; Yin, 2010), in this study we determine the age of the India–Asia collision by the paleolatitudinal intersection of the northern margin of the Himalayan terrane and the southern margin of the Lhasa terrane (Asia).

For the Lhasa terrane, the mentioned paleomagnetic results above show that the southern margin of Asia remained a relatively stable

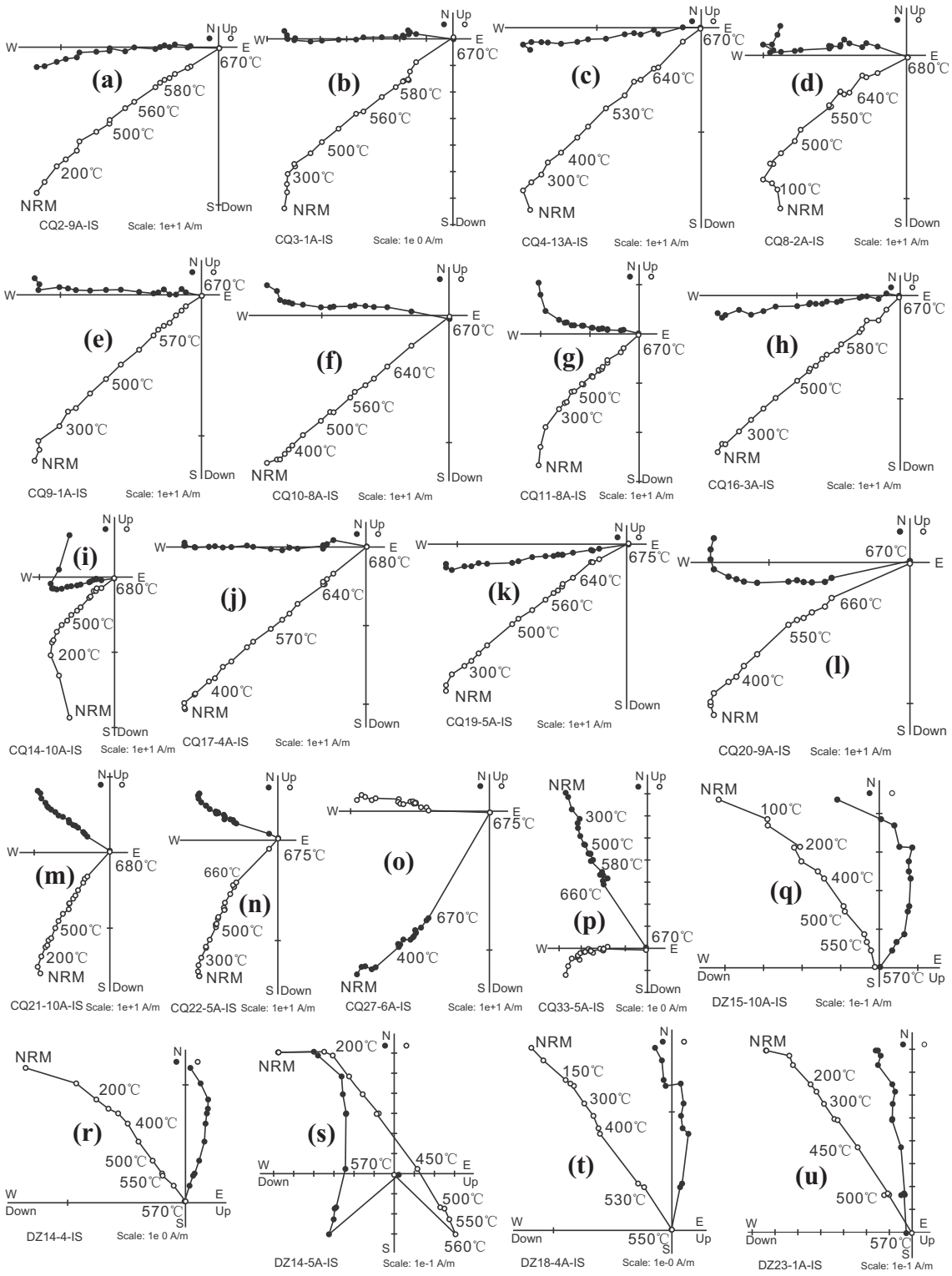


Fig. 8. Thermal demagnetization diagrams of the representative Jingzhushan Fm redbed (a–p) and Dianzhong Fm volcanic (q–u) specimens in geographic coordinates. The solid and open symbols represent the projections onto the horizontal and vertical planes, respectively.

paleolatitude during the whole Cretaceous. However, considering that an accurate paleolatitude must depend on sufficient paleomagnetic sites to fully average out paleosecular variation of the geomagnetic field (e.g., Van der Voo, 1990; Tauxe, 1993), the compaction-induced inclination shallowing is insignificant in the Upper Cretaceous redbeds of the Lhasa terrane (see Section 5.1) and that some sampling Paleocene–Eocene Linzizong volcanic rocks were erupted after the India–Asia collision, we use the paleolatitude of 15.1° ± 1.3°N (for a reference point in Zhongba county (29.77°N, 84.03°E) where it is just located in the present-day Indus–Tsangpo suture) deduced from the inclination-only mean for all the 164 (42 volcanic and 122 redbed sites) reliable Late Cretaceous paleomagnetic sites as the pre-collisional paleolatitude for the southern margin of the Lhasa terrane (Fig. 13).

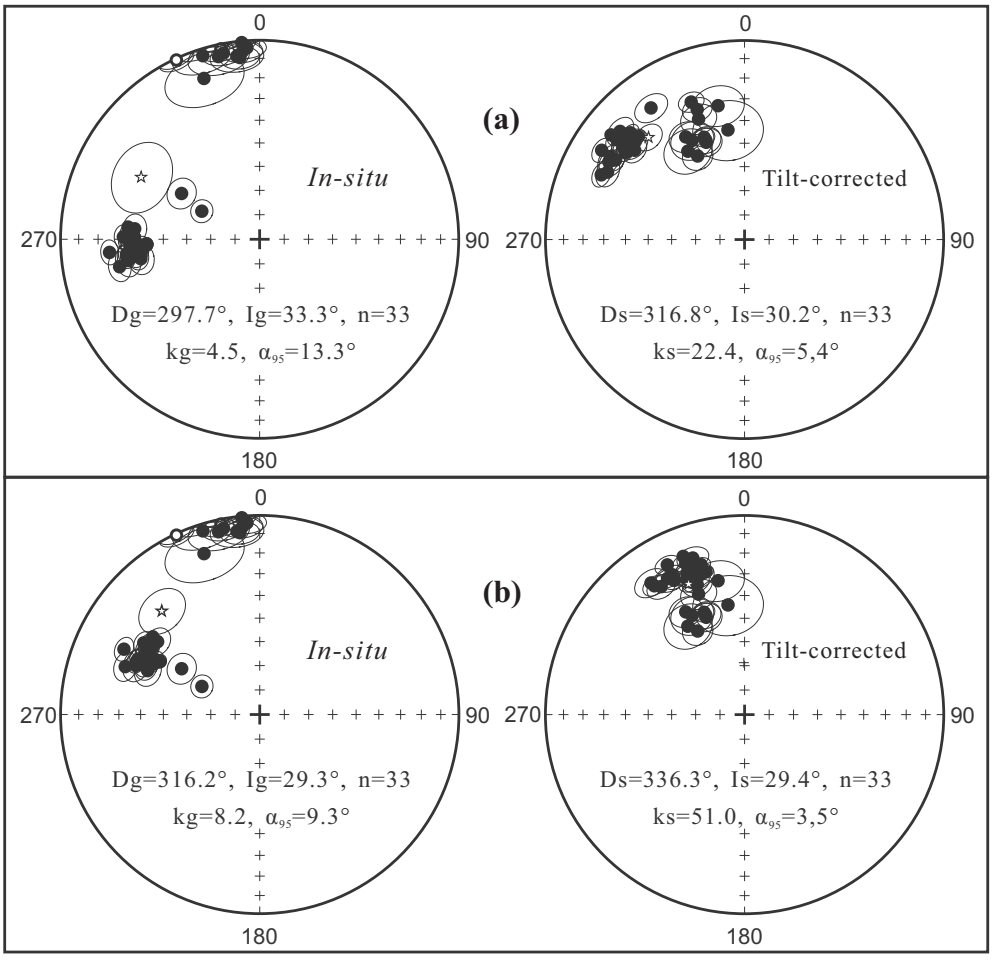


Fig. 9. Equal-area projections of site-mean directions from the Upper Cretaceous Jingzhushan Fm redbeds (a) and after a 30° clockwise rotation of the sites CQ1–20 (b). The stars indicate the overall-mean direction of 33 sites.

For the Himalayan terrane, although paleomagnetic results from the Dingri area suffered from serious remagnetization (e.g., Tong et al., 2008; Liebke et al., 2013), some from Gamba and Duela areas yielded reliable paleomagnetic data. Patzelt et al. (1996) obtained reliable paleomagnetic results, which pass positive fold tests and include dual polarity HTC directions, in the Gamba and Duela areas (Table 3) and yield paleolatitudes of 5.8° ± 3.8°N for the Zongpu Fm (55–63 Ma) and 4.4° ± 4.4°S for the Zongshan Fm (65–71 Ma) for the reference point (29.77°N, 84.03°E), respectively (Table 3). The Elongation/Inclination (E/I) analysis suggested that these limestone paleomagnetic data did not suffer from significant inclination shallowing (Dupont-Nivet et al., 2010). Recently, Yi et al. (2011) carried out a high-

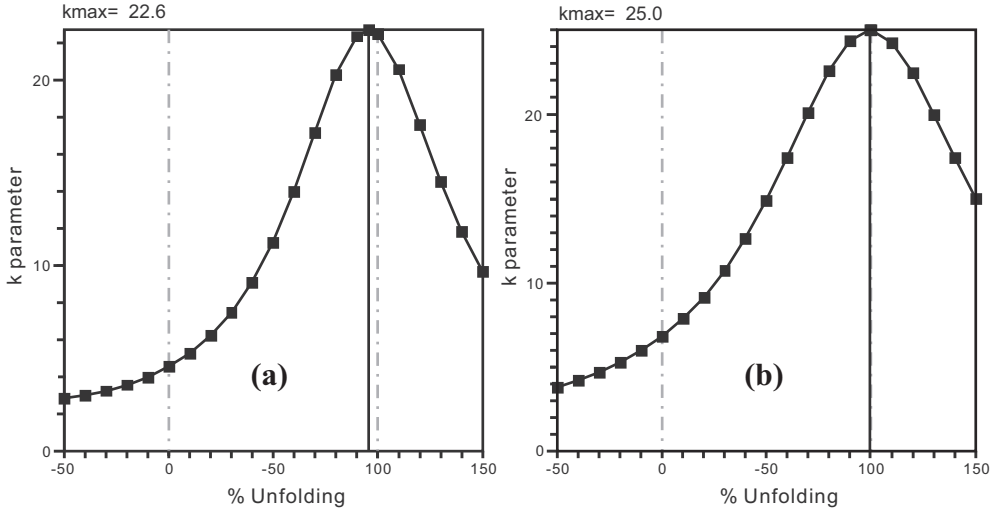


Fig. 10. Results of stepwise unfolding of the mean direction of the HTC from the Upper Cretaceous Jingzhushan Fm redbeds (a) and from the Lower Cretaceous Dianzhong Fm volcanic rocks combined with 18 previous paleomagnetic sites from the Lower Cretaceous Zenong Gp volcanic rocks reported by Chen et al. (2012).

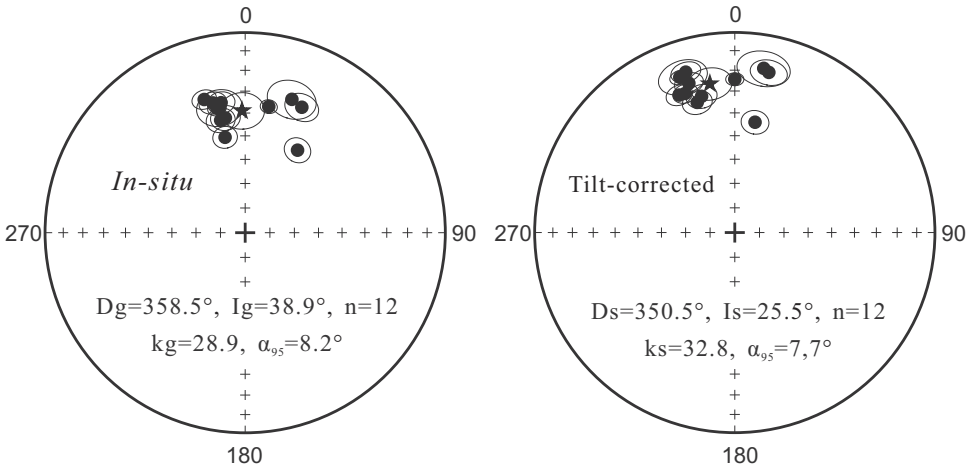


Fig. 11. Equal-area projections of site-mean directions from the Lower Cretaceous Dianzhong Fm volcanic rocks. The stars indicate the overall-mean direction of 12 sites.

resolution magnetostratigraphic investigation on the Zongpu Fm limestone in the Gamba area. Their paleomagnetic data provide two high-quality paleopoles that yield paleolatitudes of 11.8° ± 2.5°N for the time interval 59–56 Ma and 7.1° ± 3.5°N for the time interval 62–59 Ma, respectively (Table 3). To obtain a more accurate paleolatitude for the Himalayan terrane, we also have calculated a paleolatitude of 8.1° ± 1.6°N from an inclination-only mean for all the 46 Zongpu Fm (59 ± 4 Ma) limestone sites. This paleolatitude, which is very consistent with the Fisher mean paleolatitude of 8.0° ± 2.4°N, has been used as a credible paleolatitude for constraining the northern margin of the Himalayan terrane at 59 ± 4 Ma.

The northern margin of the Himalayan terrane located at 8.1° ± 1.6°N at 59 ± 4 Ma is clearly farther south than the southern margin of the Lhasa terrane located at 15.1° ± 1.3°N, indicating that the collision between the Lhasa and Himalayan terranes could not have occurred at 59 ± 4 Ma (Fig. 13). The Himalayan terrane must move ~7.0° farther north before reaching the southern margin of the Lhasa terrane. Reliable paleomagnetic results show that the Himalayan terrane at 68 ± 3 Ma and 59 ± 4 Ma was located at 4.4° ± 4.4°S and 8.1° ± 1.6°N, respectively (Table 3), revealing that the Himalayan terrane moved northward at a velocity of ~15.4 cm/yr prior to the collision. This velocity is similar to the Indian northward drift velocity of ~16 cm/yr during ~70–55 Ma derived from global plate circuits and the Indian apparent polar wander path (APWP) (Klootwijk et al., 1992; Besse and Courtillot, 2002; Torsvik et al., 2008; Molnar and Stock, 2009; Copley et al., 2010). Extrapolating the constant northward drift velocity of 15.4 cm/yr for the Himalayan terrane, it would take ~5.0 Myr (after ~59 ± 4 Ma) for its northern margin to overlap with the southern margin of the Lhasa terrane at 15.1° ± 1.3°N (Fig. 13). This provides the most likely age for the Lhasa and Himalayan collision at 54.0 ± 2.1 Ma, which corresponds with: (1) a distinct reduction in the velocity of northward drift of the Indian craton occurring at 55–50 Ma, which is generally considered as an evidence for the India–Asia collision (e.g., Klootwijk et al., 1992; Copley et al., 2010), (2) the youngest marine sediments in the Himalayan terrane dated at 54–50 Ma, which can provide a minimum age for the Lhasa and Himalayan collision since marine conditions may persist on continental crust for some time post collision (Zhu et al., 2005; Najman et al., 2010), (3) the Asian detritus on the Himalayan terrane dated at 50.6 Ma, which constrains the collision between the Lhasa and Himalayan terranes prior to 50.6 Ma (Najman et al., 2010), and (4) the ultrahigh-pressure rock formation and exhumation currently exposed in the Himalaya terrane dated at 57–50 Ma, which was considered as the initial collision age (Guillot et al., 2003, 2008).

Based on the most recent global synthetic Indian APWP (Torsvik et al., 2012) and interpolating a constant northward drift velocity between 60 (10.3° ± 2.1°S) and 50 Ma (5.5° ± 2.8°N), the location of the Indian craton at 54.0 Ma can be calculated to locate ~0.8°S for a reference point at 29.77°N, 84.03°E. Therefore, when the Himalayan terrane collided with the Lhasa terrane at ~15.1°N and ~54.0 Ma, the northern margin of the Indian craton was clearly much farther south than the southern margin of the Himalayan terrane (Fig. 13). A paleolatitudinal difference of ~15.9° (~1800 km) suggests that an intra-oceanic basin existed between the Himalayan terrane and the Indian craton during the Cretaceous and Paleocene (e.g., Ali and Aitchison, 2008; van Hinsbergen et al., 2012; Metcalfe, 2013), or the Greater India at ~54.0 Ma had a large northern extent of ~1800 km (~15.9°) beyond its present northern margin (e.g., Yi et al., 2011; Meng et al., 2012). Interpolating a constant Indian northward velocity of 11.1 cm/yr, which is deduced from the Indian APWP between 50 (5.5° ± 2.8°N) and 40 Ma (15.5° ± 2.9°N) (Torsvik et al., 2012), it would take ~9.6 Myr (after 50 Ma) for the northern margin of the Indian craton to reach the southern margin of the Himalaya terrane. This yields an age of 40.4 ± 4.1 Ma for the collision of the Indian craton and the Himalayan terrane (Fig. 13), which matches with: (1) the further slowdown of the northward drift velocity of the Indian craton occurred at ~40 Ma (Torsvik et al., 2012 and Fig. 13), and (2) the new zircon SHRIMP U–Pb dating of the leucogranite in the Tethyan Himalaya indicating that the earliest emplacement occurred at ~44–32 Ma (Qi et al., 2008; Zeng et al., 2009; Zhang et al., 2012), which was considered to represent the initial activation age of the STDS (Southern Tibetan Detachment System) and MCT (Main Central Thrust) (Xu et al., 2011).

Noticeably, Klootwijk and Bingham (1980) obtained the Early Cretaceous paleomagnetic results from the Himalayan terrane, and provided a paleopole located at 12.0°N, 289.0°E for ~118 Ma (Table 3). This Early Cretaceous pole yields a paleolatitude of 41.8° ± 6.0°S for a reference point at 29.77°N, 84.03°E, which is similar with 38.9° ± 3.3°S for 110 Ma and 40.6° ± 2.6°S for 120 Ma calculated from the Indian APWP (Table 3 and Fig. 13), supporting that the Himalayan terrane and the Indian craton were both entities before 120 Ma and the northern extension of Greater India was not more than 1000 km during the Early Cretaceous (e.g., Ali and Aitchison, 2005, 2008; van Hinsbergen et al., 2012). However, the reliable Late Cretaceous pole for the Zongshan Fm confirms that the northern margin of the Himalayan terrane at 68 ± 3 Ma was located at 4.4° ± 4.4°S for the reference point (29.77°N, 84.03°E), whereas at this time the northern margin of the Indian craton lied at 19.7° ± 3.3°S based on interpolating a constant northward drift velocity between 70 (22.0° ± 2.5°S) and 60 Ma (10.3° ± 2.1°S) (Table 3). This implies that the Himalayan terrane during ~118–68 Ma moved northward at a mean velocity of ~8.3 cm/yr, which is clearly faster than the Indian northward drift velocity of

Table 3
Summary of Cretaceous and Paleocene–Eocene paleopoles from the Lhasa and Himalayan terranes, as well as from India and East Asia reference poles.

ID	Lithology	Slat, Slon (°N, °E)	Age (Ma)	Plat (°N)	Plon (°E)	A_{95} (dp/dm) (°)	Paleolat (°N)	n/N	Test	Reference
Lhasa terrane										
QL	volc	31.7, 91.0	~90	74.0	318.0	11.1/19.1	19.7 ± 11.1	20/4	No	Lin and Watts (1988)
NQ	volc	31.5, 92.0	~96	78.0	282.0	4.0/6.9	18.3 ± 4.0	33/9	No	Lin and Watts (1988)
WR	Tuff	30.5, 90.1	~114	66.4	220.3	5.1/9.3	11.8 ± 5.1	88/15	No	Sun et al. (2008)
SX	Lava	29.9, 91.2	K2	69.1	191.7	3.3/5.4	21.7 ± 3.3	132/21	F	Tan et al. (2010)
ZN	Lava, tuff	31.4, 85.1	~110–130	58.2	341.9	4.6	19.0 ± 4.6	162/18	F, R	Chen et al. (2012)
LZ	Volc	30.5, 85.0	93–99	63.1	224.6	5.1	8.0 ± 5.1	112/14	F	Tang et al. (2013)
DZ	Volc	31.1, 84.4	117–121	70.5	292.9	7.4	12.4 ± 7.4	116/12	D	This study
XS	Redbeds	29.9, 90.7	K2	71.9	327.2	4.5	20.5 ± 4.5	111/17	F	Sun et al. (2012)
TX	Redbeds	29.9, 91.2	K2	70.2	300.5	1.4/2.7	13.3 ± 1.4	377/43	F, D	Tan et al. (2010)
PJ	Redbeds	29.9, 91.2	K2	68.0	340.0	6.7/11.6	22.4 ± 6.7	68/7	F, D	Pozzi et al. (1982)
WM	Redbeds	29.9, 91.2	K2	64.0	348.0	5.6/9.5	24.0 ± 5.6	57/6	F	Westphal et al. (1983)
TL	Redbeds	29.9, 91.2	K2	68.0	279.0	3.5/6.9	8.4 ± 3.5	51/8	F	Lin and Watts (1988)
AN	Redbeds	31.7, 91.5	K2	63.5	325.4	6.5	15.0 ± 6.5	49/6	F	Achache et al. (1984)
AS	Redbeds	29.9, 91.1	K2	71.2	288.4	7.9	12.4 ± 7.9	61/8	F	Achache et al. (1984)
[a]CQ	Redbeds	31.2, 84.7	K2	49.0	344.3	5.3	15.4 ± 2.5	291/33	F, D	This study
CY	Limestone	32.7, 80.2	K	67.7	234.2	13.1/24.5	10.0 ± 13.1	22/3	No	Chen et al. (1993)
[a]Vol_{K1}	The mean paleolatitude for the 45 Early Cretaceous volcanic sites						14.6 ± 2.4	–/42	D	This study
[a]Vol_{K2}	The mean paleolatitude for the 42 Late Cretaceous volcanic sites						15.6 ± 3.6	–/45	D	This study
[a]Vol_{K}	The mean paleolatitude for all the 87 reliable Cretaceous volcanic sites						15.1 ± 2.1	–/87	D	This study
[a]RB	The mean paleolatitude for the 122 Late Cretaceous redbed sites						14.9 ± 1.3	–/122	D	This study
[a]K2	The mean paleolatitude for the 164 Late Cretaceous volcanic and redbed sites						15.1 ± 1.3	–/209	D	This study
AJ	volc	29.9, 91.1	~48–60	71.5	300.1	6.4/11.9	14.4 ± 6.4	46/8	F, D	Achache et al. (1984)
XM	tuff	30.1, 90.9	~55	73.6	274.3	7.3	13.6 ± 7.3	99/14	F, D	Sun et al. (2010)
PN	tuff	30.0, 91.2	40–43	87.5	81.4	4.8/7.1	32.3 ± 4.8	76/9	F	Tan et al. (2010)
CJ1	volc	29.8, 89.2 30.0, 91.2	~60–64	66.0	284.9	8.5	7.1 ± 8.5	93/15	F, D	Chen et al. (2010)
LL	Dykes	30.0, 91.1	~53	68.9	225.4	5.8/10.6	12.7 ± 5.8	68/10	F	Liebke et al. (2010)
GL	volc	30.0, 91.1	47–54	76.4	212.6	4.0/6.7	20.8 ± 4.0	195/24	F, D	Dupont-Nivet et al. (2010)
CJ2	volc, sed	29.8, 89.2 30.0, 91.2	~44–60	70.6	281.0	5.3	11.1 ± 5.3	148/23	F, D	Chen et al. (2010)
MJ	sed	29.9, 84.3	54–57	78.0	329	5.9	24.2 ± 5.9	62/–	F	Meng et al. (2012)
GS	sed, tuff	30.0, 91.2	43–54	68.4	243.0	1.9	9.4 ± 1.9	119/–	No	Huang et al. (2013)
Himalayan terrane										
PA	Limestone	28.3, 88.5 28.0, 89.2	55–63	65.4	277.6	3.8/7.6	5.8 ± 3.8	113/14	F, D	Patzelt et al. (1996)
YZ1	Limestone	28.3, 88.5	56–59	71.6	277.8	2.5	11.8 ± 2.5	141/14	F, R	Yi et al. (2011)
YZ2	Limestone	28.3, 88.5	59–62	67.3	266.3	3.5	7.1 ± 3.5	171/18	F, R	Yi et al. (2011)
[a]CP	Limestone	28.3, 88.5 28.0, 89.2	55–63	68.1	270.5	2.4	8.1 ± 1.6	425/46	F, R	This study
ZS	Limestone	28.3, 88.5 28.0, 89.2	65–71	55.8	261.4	4.4/8.6	−4.4 ± 4.4	156/14	F, D	Patzelt et al. (1996)
ZP	Sandstone	28.8, 83.8	~118	12.0	289.0	6.0/7.5	−41.8 ± 6.0	95/–	No	Klootwijk and Bingham (1980)
Indian craton										
India APWP			40	−74.7	106.8	2.9	15.5 ± 2.9			Torsvik et al. (2012)
India APWP			50	−65.1	98.4	2.8	5.5 ± 2.8			Torsvik et al. (2012)
India APWP			60	−48.5	100.8	2.1	−10.3 ± 2.1			Torsvik et al. (2012)
India APWP			70	−36.4	110.7	2.5	−22.0 ± 2.5			Torsvik et al. (2012)
East Asia										
East Asia APWP			70	79.7	219.5	3.3	22.2 ± 3.3			Cogné et al. (2013)
East Asia APWP			80	79.5	216.9	3.4	22.4 ± 3.4			Cogné et al. (2013)
East Asia APWP			90	77.9	212.8	2.9	21.8 ± 2.9			Cogné et al. (2013)
East Asia APWP			70–90	79.0	216.2	1.8	22.1 ± 1.8			This study

Notes: volc, volcanic rocks; sed, sedimentary rocks; K1, Early Cretaceous; K2, Late Cretaceous; K, Cretaceous; Slat (Slon), latitude (longitude) of sites; Plat (Plon), latitude (longitude) of poles; A_{95}, the radius that the mean pole lies within 95% confidence; dp/dm, semi-axes of elliptical error of the pole at a probability of 95%; Paleolat, paleolatitude calculated for the reference point at 29.77°N, 84.03°E; n/N, number of samples or sites used to calculate Fisher mean; F, positive fold test; R, positive reversal test; D, coexistence of dual-polarity.
[a] ID implies that the paleolatitude was obtained using the inclination-only mean calculated following the method of Arason and Levi (2010).

~4.9 cm/yr (Fig. 14). This leads to the ~1700 km separation of the Himalayan terrane from the Indian craton. Furthermore, the balanced cross-section analyses from various parts of the Himalayas suggest that an amount of 670–775 km accommodated by folding and thrusting within the Himalayas (e.g., DeCelles et al., 2002; Yin et al., 2010), which is much less than the ~1800 km northern extension of Greater India. We thus suggest that an intra-oceanic basin existed between the Himalayan terrane and the Indian craton, rather than the Greater India having a large northern extent of ~1800 km beyond its present northern margin. Although such an intra-oceanic basin still needs to be validated, based on reliable Cretaceous and Paleocene paleomagnetic results from the Lhasa and Himalayan terranes, as well as the Indian APWP, we tentatively propose that the India–Asia collision was a complex process (Figs. 13 and 14), which consists of the collision of the Lhasa and Himalayan terranes at 54.0 ± 2.1 Ma, the longtime subduction of an intra-oceanic basin between the Himalayan terrane and the Indian craton from ~54.0 to ~40.4 Ma, and the collision of the Himalayan terrane and the Indian craton at 40.4 ± 4.1 Ma. Subsequently, the continuous India–Asia continent–continent collision resulted in not only widespread intracontinental deformation between the Indian craton

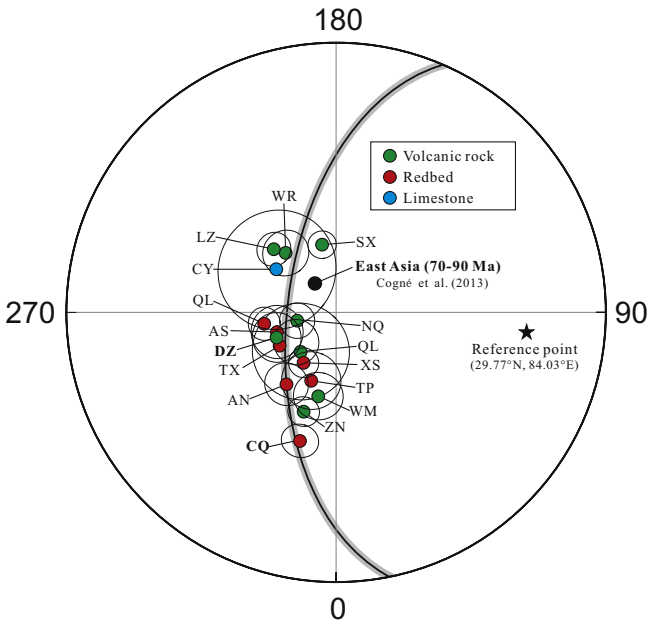


Fig. 12. Equal-area projections of the Cretaceous paleomagnetic poles of the Lhasa terrane and the Late Cretaceous (70–90 Ma) reference poles of East Asia. See Table 3 for the value, abbreviation and reference of the paleopoles. Small-circle with its 95% errors passing through a paleolatitude of 15.1° ± 1.3°N was calculated from an inclination-only mean for all the 164 reliable Late Cretaceous paleomagnetic sites, and centered in the reference point with a star at 29.77°N, 84.03°E.

and stable Siberia (e.g., Molnar and Tapponnier, 1975; Matte et al., 1996; Meyer et al., 1998), but also the uplift of the Tibetan Plateau (e.g., Chung et al., 1998).

5.3. Amount of intracontinental shortening within Asia

The Besse and Courtillot (1991) Eurasian APWP, reappraised by Besse and Courtillot (2002), is most widely employed as a reference curve for late Meso-Cenozoic paleomagnetic studies in Asia. It should be noted that paleomagnetic data used for constructing the Besse and Courtillot (2002) Eurasian APWP are based mainly from northwestern Europe, and some recent studies showed that relative tectonic movements between the western (European) and eastern (Siberian) parts of the Eurasian Plate were still active during the Cenozoic (e.g., Trifonov, 2004; Bergerat et al., 2007). Cogné et al. (2013) therefore considered that the Besse and Courtillot (2002) Eurasian APWP is not applicable to Siberia and East Asia, and constructed a new East Asia APWP using paleomagnetic data from the Siberian craton, Amuria, the North and South China Blocks and the Korean Peninsula where they formed a single entity in the Meso-Cenozoic. In this study we therefore use the new East Asia APWP as a reference for estimating intracontinental shortening between the Lhasa terrane and stable Asia (the Hexi corridor).

The Cretaceous paleomagnetic results obtained from the Lhasa terrane indicate that its southern margin was located at 15.1° ± 1.3°N before the India–Asia collision for the reference point at 29.77°N, 84.03°E (see Section 5.1). Comparing with the paleolatitude of 22.1° ± 1.8°N expected from the Late Cretaceous (70–90 Ma) average pole (79.0°N, 216.2°E with A_{95} = 1.8°) for the new East Asia APWP (Cogné et al., 2013) reveals a paleolatitude difference of 7.0° ± 2.2°, indicating that a latitudinal convergence of 780 ± 240 km has taken place between the Lhasa terrane and East Asia (the Hexi corridor) since the India–Asia collision. The 780 ± 240 km north–south shortening deduced from reliable Cretaceous paleomagnetic data is very consistent with 600–1000 km absorbed by Cenozoic fold and thrust belts between the Lhasa terrane and the Hexi corridor (NCB) (Meyer et al., 1998; Yin and Harrison, 2000; Spurlin et al., 2005; Yin et al., 2007; Lippert et al., 2011;

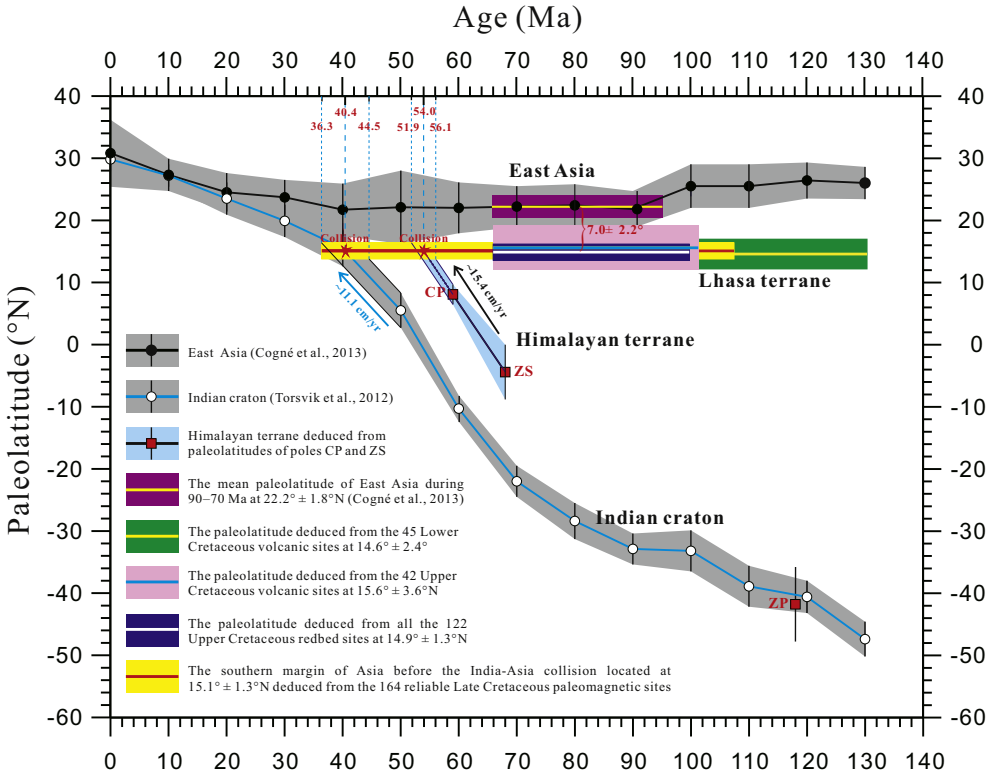


Fig. 13. Paleolatitude plots for East Asia, the Indian craton, the Himalayan terrane and the Lhasa terrane. The shaded areas and vertical bars display the errors of the paleolatitudes. The expected and observed paleolatitudes were calculated for the reference point at 29.77°N, 84.03°E.

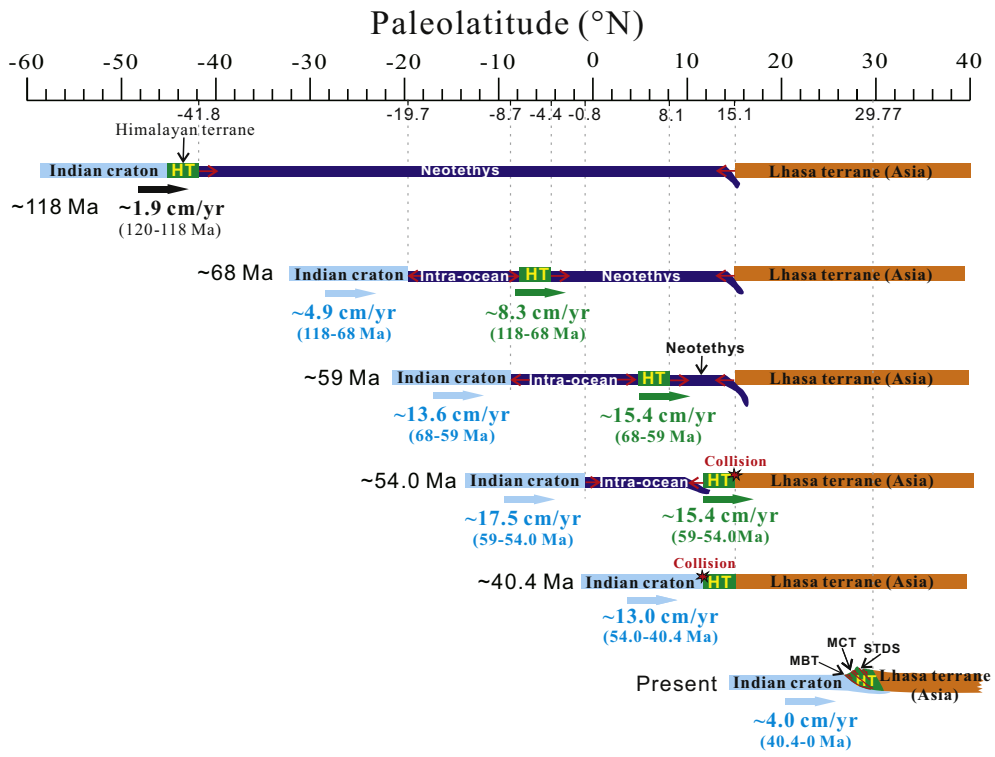


Fig. 14. A conceptional model showing the geodynamic evolution of the India–Asia collision process.

van Hinsbergen et al., 2011). These Cenozoic fold and thrust belts include the Shiquanhe–Gaize–Amdo thrust system (120–250 km), the Fenghuo Shan–Nangqian fold and thrust belt (70–110 km), the Qimen Tagh–North Kunlun thrust system (70–270 km), and the Nan Shan thrust belt (~360 km).

6. Conclusions

We have carried out a combined geochronologic and paleomagnetic study of the Upper Cretaceous Jingzhushan Fm redbeds and the Lower Cretaceous Dianzhong Fm volcanic rocks from the Cuoqin area in the central Lhasa terrane. Based on the our new geochronologic and paleomagnetic results, as well as previous paleomagnetic poles from the Lhasa and Himalayan terranes, East Asia and India craton, our conclusions can be summarized as follows:

1. Our U–Pb zircon geochronologic results from the Dianzhong Fm volcanic rocks yield the concordant ages of 117.03 ± 0.51 Ma and 121.01 ± 0.66 Ma, which are consistent with the adjacent Zenong Gp volcanic rocks. These new U–Pb zircon ages indicate that the Dianzhong Fm volcanic rocks in the Cuoqin sampling area were erupted during the Early Cretaceous, clearly different from the Dianzhong Fm volcanic rocks in the Linzhou basin has been widely accepted to have a Paleocene age.
2. Thirty-three Late Cretaceous Jingzhushan redbed sites provide a tilt-corrected site-mean direction of D = 316.8°, I = 30.2° with α_{95} = 5.4°, corresponding to a paleopole at 49.0°N, 344.3°E with A_{95} = 5.3°. Dual polarity HTCs, positive fold tests and a maximum grouping at 96% unfolding reveal that the HTCs isolated from the Jingzhushan Fm redbeds were a prefolding primary magnetization before the latest Paleocene.
3. Twelve Early Cretaceous Jingzhushan Fm volcanic sites yield a tilt-corrected site-mean direction of D = 350.5°, I = 25.5° with α_{95} = 7.7°, corresponding to a paleopole at 70.5°N, 292.9°E with A_{95} = 7.4°. The existence of dual polarity HTCs, as well as a combined site-mean direction passes positive fold tests and yields a maximum grouping at 100% unfolding, indicate that the HTCs obtained from the Dianzhong Fm volcanic rocks should be a prefolding primary magnetization before the latest Early Cretaceous.
4. For a reference point in Zhongba county (29.77°N, 84.03°E) where it is just located in the present-day Indus–Tsangpo suture, the 122 Late Cretaceous redbed sites yield an inclination-only mean of 28.0° ± 2.1° and a corresponding paleolatitude of 14.9° ± 1.3°N that are fully consistent with 29.2° ± 5.9° and 15.6° ± 3.6°N deduced from the 42 reliable Late Cretaceous volcanic sites, showing that the compaction-induced inclination shallowing is insignificant in the Upper Cretaceous redbeds of the Lhasa terrane.
5. For a reference point in Zhongba county (29.77°N, 84.03°E) where it is just located in the present-day Indus–Tsangpo suture, the southern margin of Asia was long-termly located at 15.1° ± 1.3°N before the India–Asia collision.
6. Comparing the Cretaceous–Eocene paleomagnetic results observed from the Lhasa terrane with those reported from the Indian APWP and the Himalayan terrane suggests that the India–Asia collision should be a complex process, consisting of the collision of the Lhasa and Himalayan terranes at 54.0 ± 2.1 Ma, the longtime subduction of an intra-oceanic basin between the Himalayan terrane and the Indian craton from ~54.0 to ~40.4 Ma, and the collision of the Himalayan terrane and the Indian craton at 40.4 ± 4.1 Ma.
7. Comparison with the Late Cretaceous (70–90 Ma) average pole of the East Asia APWP reveals that a latitudinal convergence of 780 ± 240 km has taken place between the Lhasa terrane and East Asia (the Hexi corridor) since the India–Asia collision. The shortening amount estimated from paleomagnetic data is fully consistent with the 600–1000 km accommodated by the Cenozoic fold and thrust belts between the Lhasa terrane and Hexi corridor.
8. The Cretaceous paleomagnetic results observed from the Lhasa terrane support that local vertical-axis rotations have taken place after the Cretaceous.

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